

Marbovitryl FD 5 mg TABLET

Marbofloxacin 5 mg tablets

COMPOSITION: Each tablet contains:
Marbofloxacin...5.0 mg

PRODUCT DESCRIPTION
Beige, oval shaped, scored, uncoated tablets

PHARMACOLOGICAL PROPERTIES
Pharmacotherapeutic group: Antibacterials for systemic use;
Fluoroquinolones; Marbofloxacin.
ATC vet code: QJ01MA93

Pharmacodynamic properties
Marbofloxacin is a synthetic, bactericidal antimicrobial, belonging to the fluoroquinolone group which acts by inhibition of DNA gyrase. It is effective against a wide range of Gram positive bacteria (in particular *Staphylococci*, *Streptococci*) and Gram negative bacteria (*Escherichia coli*, *Salmonella typhimurium*, *Citrobacter freundii*, *Enterobacter cloacae*, *Serratia marcescens*, *Morganella morganii*, *Proteus spp.*, *Klebsiella spp.*, *Shigella spp.*, *Pasteurella spp.*, *Haemophilus spp.*, *Moraxella spp.*, *Pseudomonas spp.*, *Brucella canis*) as well as *Mycoplasma spp.*

Bacterial strains with a MIC $\leq$ 1 mcg/ml are susceptible, strains with a MIC of 2 mcg/ml are intermediately susceptible and strains with a MIC $\geq$ 4 mcg/ml are resistant to marbofloxacin (CLSI, 2004).

Resistance to fluoroquinolones occurs mostly by chromosomal mutation with three mechanisms: decrease of the bacterial wall permeability, expression of efflux pump or mutation of enzymes responsible for molecule binding.

Marbofloxacin is not active against anaerobes, yeasts or fungi.

Pharmacokinetic particulars
After oral administration in dogs and cats at the recommended dose of 2 mg/kg, marbofloxacin is readily absorbed and reaches maximal plasma concentrations of 1.5 µg/ml within 2 hours.

Its bioavailability is close to 100%. It is weakly bound to plasma proteins (less than 10%), extensively distributed and in most tissues (liver, kidney, skin, lung, bladder, digestive tract) it achieves higher concentrations than in plasma. Marbofloxacin is eliminated slowly ($t_{1/2}$ = 14 h in dogs and 10h in cats) predominantly in the active form in urine (2/3) and faeces (1/3).

INDICATION
Marbofloxacin is indicated in the treatment of the following infections caused by susceptible strains of organisms;

Dogs
- Skin and soft tissue infections (skinfold pyoderma, impetigo, folliculitis, furunculosis, cellulitis)
- Urinary tract infections (UTI) associated or not with prostatitis or epididymitis
- Respiratory tract infections
Cats
- Skin and soft tissue infections (wounds, abscesses, phlegmons)
- Upper respiratory tract infections

TARGET SPECIES: Dogs and Cats

ROUTE OF ADMINISTRATION: For oral use

RECOMMENDED DOSE: For oral administration.
The recommended dose rate is 2 mg/kg/day in a single daily administration (see table below).

Body Weight	Tablets
1.3 - 2.5 kg	1
2.6 - 5.0 kg	2

To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.

Dogs:
- In skin and soft tissue infections, treatment duration is at least 5 days. Depending on the course of the disease, it may be extended up to 28 days.
- In urinary tract infections, treatment duration is at least 10 days. Depending on the course of the disease, it may be extended up to 40 days.
- In respiratory infections, treatment duration is at least 7 days and depending on the course of the disease, it may be extended up to 21 days.

Cats:
- For skin and soft tissue infections (wounds, abscesses, phlegmons) treatment duration is 3 to 5 days.
- For upper respiratory infections treatment duration is 5 days.

CONTRAINDICATION
Do not use in dogs aged less than 12 months, or less than 18 months for exceptionally large breeds of dogs, such as Great Danes, Briard, Bernese, Bouvier and Mastiffs, with a longer growth period.

Do not use in cats aged less than 16 weeks,

Do not use in animals with known hypersensitivity to marbofloxacin or other (fluoro)quinolones or to any of the excipients (Calcium hydrogen phosphate, Microcrystalline Cellulose, Sodium Starch Glycolate, Carboxymethylcellulose, Magnesium stearate).
Do not use in cases of confirmed or suspected resistance to fluoroquinolones (cross resistance).

INTERACTION WITH OTHER MEDICAL PRODUCT AND OTHER FORMS OF INTERACTION
Fluoroquinolones are known to interact with orally administered cations (Aluminium, Calcium, Magnesium, Iron). In such cases, the bioavailability may be reduced.
When administered together with theophylline, the half-life and thus the plasma concentration of theophylline increase. Hence, in case of concurrent administration the dose of theophylline should be reduced.
Do not use in combination with tetracyclines, macrolides because of the potential antagonist effect.

WARNING AND PRECAUTION

- Keep out of reach of children
- Jauhkan daripada kanak-kanak
Special precautions for use in animals
The fluoroquinolones have been shown to induce erosion of articular cartilage in juvenile dogs and care should be taken to dose accurately especially in young animals. However, at the therapeutic recommended dosage, no severe side effects are to be expected in dogs and cats.

Some fluoroquinolones at high doses may have an epileptogenic potential. Cautious use is recommended in dogs and cats diagnosed as suffering from epilepsy.
Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly to other classes of antimicrobials. Whenever possible, use of fluoroquinolones should be based on susceptibility testing. Use of the product deviating from the instructions given in the SPC may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease effectiveness of treatment with other quinolones due to the potential for cross-resistance. Official and local antimicrobial policies should be taken into account when the product is used.
Special precautions to be taken by the person administering the veterinary medicinal product to animals
People with known hypersensitivity to (fluoro)quinolones should avoid using this product. Avoid contact of the skin and eyes with the product. In case of accidental ingestion seek

medical advice immediately and show the package leaflet or the label to the physician. Wash hands after use.

SPECIAL WARNINGS FOR EACH TARGET SPECIES
A low urinary pH could have an inhibitory effect on the activity of marbofloxacin.

ADVERSE REACTIONS

Mild side effects such as vomiting, softening of faeces, modification of thirst or transient increase in activity may very rarely occur. These signs cease spontaneously after treatment and do not necessitate cessation of treatment.
The frequency of adverse reactions is defined using the following convention:
- very common (more than 1 in 10 animals treated displaying adverse reaction(s))
- common (more than 1 but less than 10 animals in 100 animals treated)
- uncommon (more than 1 but less than 10 animals in 1,000 animals treated)
- rare (more than 1 but less than 10 animals in 10,000 animals treated)
- very rare (less than 1 animal in 10,000 animals treated, including isolated reports).

USE DURING PREGNANCY, LACTATION OR LAY
Studies in laboratory animals (rats, rabbits) showed no teratogenicity, embryotoxicity and maternotoxicity with marbofloxacin at therapeutic doses.
The safety of marbofloxacin has not been assessed in pregnant and lactating cats and dogs. Use only accordingly to the benefit/risk assessment by the responsible veterinarian in pregnant and lactating animals.

OVERDOSE
Overdosage may cause acute signs in the form of neurological disorders, which should be treated symptomatically

WITHDRAWAL PERIOD: Not applicable.

STORAGE CONDITION

Store in a cool dry place not exceeding 30°C.

SHELF LIFE: 2 years
SHELF LIFE AFTER OPENING: 24 hours
PACKAGING: Box contains 10 blisters of 10 tablets (100 tablets)
REVISED DATE: February 07, 2023

Vemedim

Manufactured by:
Vemedim Veterinary Drugs and Probiotics Co. Ltd
(Vemedim Animal Health)
Song Hau Industrial Zone, Dong Phu Commune,
Chau Thanh District, Hau Giang Province, Vietnam

Product register holder. Distributed by:
Vemedim Sdn Bhd
15, Jalan USJ 1/31, 47600 Subang Jaya
Selangor, Darul Ehsan, Malaysia
Tel: (603)80247118 Fax: (603) 80247168